

2.2 Evaluating the Clarity of an Algorithm

وضاحت (Clarity) کا مطلب ہے الگورتھم پڑھنا، سمجھنا اور بدلنا آسان ہو۔

CS XII

Chapter 2: Computational Thinking & Algorithms

Lecture 03 of 16

2.2 Evaluating the Clarity of an Algorithm

LEARN IT

Clarity means an algorithm is easy to **read, understand and modify**. In many cases, evaluating clarity is even more important than making the algorithm slightly faster. A clear algorithm is easier to **maintain, debug and extend** over time.

The two key aspects of clarity are:

Clarity → Modularity (parts) → Readability (text and flow)

WORKED EXAMPLE

Worked example: speed versus clarity. Algorithm A finds the class average in 8 confusing lines with variables a , b , c . Algorithm B uses two named functions and takes the same number of additions. Both are $O(n)$. Algorithm B is the better school solution because the next person (or you, in three months) can change it without breaking it.

REMEMBER IT

Ask two questions: **"Can I split this into parts?"** (modularity) and **"Can another student read it aloud and understand it?"** (readability).

COMMON MISTAKE

Mistake: treating a clever one-line trick as "better" than a clear five-line solution.

Fix: the source says clarity is often more important than speed. Board answers should praise meaningful names and modules, not compacted code.

اردو میں سمجھیں

وضاحت (Clarity) کا مطلب ہے الگورتھم پڑھنا، سمجھنا اور بدلنا آسان ہو۔ بہت سی صورتوں میں وضاحت رفتار سے بھی اہم ہے۔ اس کے دو حصے ہیں: ماڈیولریٹی (Modularity) یعنی کام کو حصوں میں بانٹنا، اور پڑھائی پن (Readability) یعنی متن اور بہاؤ صاف ہونا۔

CHECK YOURSELF

1. What does clarity mean?
2. Name the two key aspects of clarity.
3. Why can clarity matter more than a small speed gain?

Answers: 1. The algorithm is easy to read, understand and modify. 2. Modularity and readability. 3. Because a clear algorithm is easier to maintain, debug and extend over time.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Clarity means the algorithm is:

- a. Easy to read, understand, change
- b. As short as possible even if cryptic
- c. Written in one letter variables only
- d. Uncommented hex

MCQ 2. Modularity helps clarity because:

- a. Named pieces have one job
- b. Everything is in main
- c. There are no names
- d. Gotos everywhere

MCQ 3. Readability includes:

- a. Names, structure, comments, consistent style
- b. Removing all spaces
- c. 1-letter names always
- d. No line breaks

MCQ 4. A 200-line monoblock is:

- a. Low modularity
- b. High modularity
- c. A primary key
- d. A pie

MCQ 5. Clarity can matter more than a tiny speed gain when:

- a. Humans must maintain the code
- b. $n=10^{12}$ always
- c. It is a one-off NAND
- d. It is OSI layer 1

MCQ 6. Poor names are a:

- a. Readability problem
- b. TCP problem
- c. Power problem
- d. RAM manufacturing problem

Extra short questions

1. Define clarity of an algorithm.
2. Two ways to improve it.
3. Can a correct algorithm be unclear?
4. Link to functions in Ch 3.

Extra long question

1. Rewrite a messy "average of array" monoblock as three modules (input, compute, output). Explain the clarity gain.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. A person can see the purpose and logic without running it.
2. Decompose into modules; meaningful names; comments on why; consistent structure.
3. Yes — and then it will be modified wrongly.
4. Functions are modularity in Python.

Long

1. Show before/after pseudocode. Each module tested. Names: `sum_list`, `mean`.
Comments why `n=0` is rejected.

Next lecture: 2.2.1 – 2.2.2 Modularity and Readability